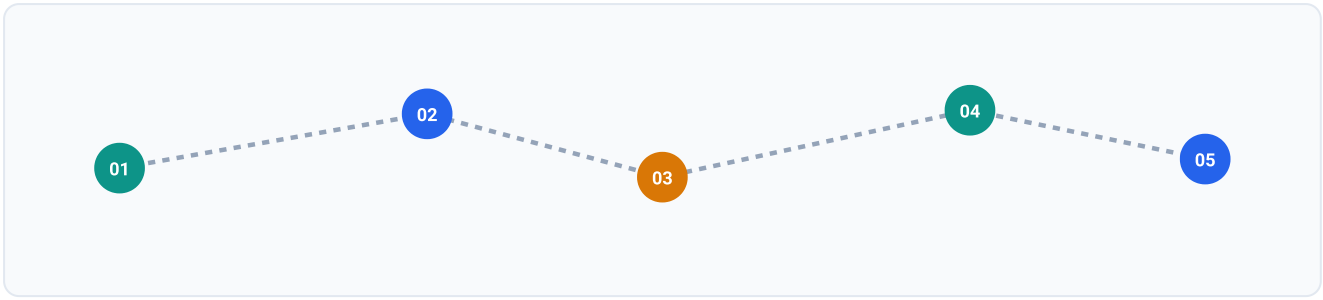
AIORI-3 HACKATHON • TECHNICAL PROPOSAL

PopPILOT INTERNET-QUALITY COMPARISON BY VERIFIED ACCESS MEDIUM

"Measure the connection, not just the speed."



TEAM CODE
A3-PS014-TC182

PROBLEM STATEMENT
PS-014

MEASUREMENT DIMENSIONS

01 LATENCY
Round-trip time to defined targets

02 PACKET LOSS
Probe failure percentage

03 DNS DELAY
Hostname resolution time

04 THROUGHPUT
Bounded transfer rate

TEAM MEMBERS
Abhishek Giri
Ansh Kapoor

FACULTY MENTOR
Er. Mukesh Verma

A repeatable pilot with transparent context, controlled probes, and cautious comparisons. **Proposed design. No pilot outcomes are claimed before measurements are collected.**

Contents & Project Snapshot

A quick guide to the proposal and the evidence the pilot is designed to produce.

NO.	SECTION	PURPOSE
01	Project overview	Purpose, scope and pilot parameters
02	Contents & snapshot	Document map and scope
03	Problem description	Limits of one-off speed tests
04	Proposed solution	Scheduled probes and context-aware records
05	Proposed optimizations	Repeatability, quality controls and fair comparison
06	Architecture & data design	Collection, ingestion, storage and dashboard
07	Delivery timeline	Build stages and seven-day pilot
08	References	Primary methodology and official resources
09	Conclusion	Expected outputs, boundaries and validation

PILOT SCALE

10-20 Volunteer Endpoints

A small, manageable pilot population.

MEDIA COVERAGE

≥ 2 Access Media

Labels retain their evidence source.

DURATION

7 Scheduled Days

Repeated observations, not one-off tests.

PROJECT THESIS

Internet quality cannot be summarized by peak download speed alone. The proposed platform pairs repeated network measurements with access-medium provenance, ISP/location context and local-link notes so that observed differences can be interpreted more carefully.

CORE MEASUREMENTS SUMMARY

LATENCY
Round-trip time to a defined target, recorded with timestamp and test status.

PACKET LOSS
Unanswered probes reported as a percentage, with failed runs preserved.

DNS DELAY
Time needed to resolve the defined test hostname.

LIMITED THROUGHPUT
A bounded transfer test against a controlled endpoint.

Problem Description

Peak bandwidth is only one dimension of the experience people get from an Internet connection.

A speed test can report a strong download rate while users still experience slow page starts, delayed responses or unreliable sessions. Those symptoms may relate to latency, packet loss or DNS resolution delay, and can change over time. Results also reflect local conditions, including Wi-Fi, device, ISP and location. A useful comparison therefore needs consistent probes and enough context to interpret the observations.

01 SINGLE-SNAPSHOT BIAS

One result cannot show whether quality remains stable across different times.

02 BANDWIDTH-ONLY VIEW

Peak throughput does not describe latency, packet loss or DNS lookup delay.

03 DIFFERENT ACCESS MEDIA

Fiber, cellular, fixed wireless and satellite operate under different conditions.

04 LOCAL-LINK CONFOUNDERS

Wi-Fi state, device configuration, ISP and location can influence measurements.

05 UNCERTAIN LABELS

Access-medium claims need a recorded source, not an assumption based on IP alone.

06 SMALL-SAMPLE LIMITS

A volunteer pilot can reveal patterns, but cannot establish universal or causal rankings.

WHY CONTEXT MATTERS

OBSERVED FACTOR	HOW THE PROPOSAL HANDLES IT
Access medium	Store the label and its source; do not infer it from IP address alone.
Wi-Fi/local link	Capture context and, where feasible, gateway RTT as a reference signal.
ISP and location	Retain coarse region and ISP label to help interpret differences.
Missing or failed tests	Record failures and sample counts instead of silently excluding them.

SCOPE BOUNDARY

This is a controlled pilot, not a national coverage study. It will compare observed trends with context and uncertainty, without claiming perfect isolation of Wi-Fi and ISP effects or a universal winner.

Proposed Solution

A scheduled measurement pipeline that treats context and provenance as part of the result.

The system coordinates repeated probes from volunteer endpoints to a controlled measurement destination. Each run produces a timestamped record with the core metrics, status information and context fields needed for later comparison. The dashboard presents trends and distributions instead of collapsing all dimensions into a single score.

01 COLLECT

A lightweight agent runs the same defined probes on a schedule and records successful, failed or skipped runs.

02 ENRICH

Attach access-medium label/source, ISP, coarse region, Wi-Fi context, timestamp and agent version.

03 INTERPRET

Summarize sample counts, median/percentiles, variation and uncertainty when supported by the data.

END-TO-END WORKFLOW

STEP 1
Volunteer Endpoint
Consent + context

STEP 2
Scheduler
Defined test cadence

STEP 3
Measurement Agent
Run core probes

STEP 4
API + Database
Validate + store

STEP 5
Analysis Layer
Sample quality + spread

STEP 6
Dashboard
Filters + trends

WHAT MAKES THE APPROACH DIFFERENT

PROVENANCE BY DESIGN

The medium label is stored with its source so reviewers can distinguish verified evidence from participant-reported context.

TRANSPARENT UNCERTAINTY

The dashboard keeps sample counts and spread visible; it does not hide weak samples behind a single quality score.

No AI classifier, automatic medium detection or formal RFC certification is claimed. The contribution is a reproducible measurement process and a more careful interpretation of the resulting data.

Proposed Optimizations

Practical improvements that strengthen repeatability, data quality and fairness.

REPEATABLE SCHEDULING

Approach: Run probes at a defined cadence and use the same destination and procedure across endpoints.

Benefit: Supports time-trend analysis instead of reliance on one snapshot.

LABEL PROVENANCE

Approach: Store the access-medium label, evidence source and relevant ISP/location context.

Benefit: Makes assumptions explicit and comparison limitations easier to review.

BOUNDED THROUGHPUT

Approach: Use a fixed-size transfer, explicit time limit and controlled endpoint.

Benefit: Reduces resource consumption and makes test conditions more consistent.

FAILURE-AWARE DATA

Approach: Keep timeout, error, skipped-run and agent-version fields in the dataset.

Benefit: Stops failed tests from disappearing and distorting sample counts.

DISTRIBUTION-FIRST DASHBOARD

Approach: Report sample count, median, suitable percentiles and variation where sample size permits.

Benefit: Avoids false precision from a lone average or invented composite score.

LOCAL-LINK REFERENCE

Approach: Record gateway RTT where feasible, beside controlled-target RTT.

Benefit: May flag a local-link issue; it does not perfectly separate Wi-Fi, ISP and access-medium effects.

COMPARISON RULE

Do not label a medium inherently superior based on this pilot alone. Interpret observed differences with endpoint counts, test failures, ISP/location context, Wi-Fi notes and the spread of measurements.

Loaded jitter is a possible extension only if its measurement method, units and validation are implemented. The required core remains latency, packet loss, DNS delay and limited throughput.

Architecture & Data Desig

A modular design makes each stage testable and keeps raw observations traceable.

01 COLLECT
- Volunteer endpoints
- Consent and context
- Medium label + source
- ISP & coarse region

02 MEASURE
- Scheduler / agent
- Latency & loss
- DNS timing
- Bounded transfer

03 INGEST + STORE
- FastAPI validation
- Timestamped records
- Error & test status
- SQLite / PostgreSQL

04 ANALYSE + PRESENT
- Sample counts
- Median & percentiles
- Spread & uncertainty
- Dashboard & report

PROPOSED DATA RECORD SCHEMA

FIELD GROUP	EXAMPLE FIELDS	PURPOSE
Identity + status	`measurement_id`, `participant_id`, `timestamp`, `test_status`	Trace each run without unnecessary personal data.
Medium + context	`access_medium`, `label_source`, `ISP`, `coarse_region`, `wifi_context`	Record comparison context and evidence provenance.
Core metrics	`latency_ms`, `packet_loss_pct`, `dns_delay_ms`, `throughput_mbps`	Store measurements with explicit units.
Reproducibility	`agent_version`, `target_id`, `probe_configuration`	Help reproduce or explain differences between runs.

PROPOSED STACK

Python measurement agent • FastAPI backend • SQLite for initial pilot (PostgreSQL if needed) • React dashboard and charts.

PRIVACY + DATA QUALITY

Use pseudonymous IDs, coarse location and explicit failure records. Avoid collecting unnecessary identifiers.

This is the proposed architecture. The submission should only describe components as implemented after they are present in the repository and can be demonstrated.

Delivery Timeline & Acceptance

A staged plan distinguishes implementation work from the actual seven-day observation period.

STAGE 01 / SPECIFY
Define probe procedures, target, data fields and participant context.

STAGE 02 / BUILD
Implement measurement agent, API ingestion and validation.

STAGE 03 / STORE + ANALYSE
Connect storage, quality checks, summaries and dashboard filters.

STAGE 04 / TEST
Exercise timeouts, failed probes, units, missing records and repeatability.

STAGE 05 / ONBOARD
Recruit 10-20 endpoints and confirm at least two access media.

STAGE 06-12 / PILOT
Run scheduled measurements for seven days and preserve logs.

STAGE AFTER PILOT / REPORT
Summarize actual observations, sample counts, uncertainty and limitations.

DELIVERABLE ACCEPTANCE CHECKS

DELIVERABLE	EVIDENCE TO SHOW
Measurement agent	Timestamped metric output plus explicit error and test-status records.
Controlled endpoint	Defined destination and documented probe procedure shared across endpoints.
Labelled dataset	Access-medium source and context retained with each usable measurement.
Dashboard + report	Filters, sample counts, distribution summaries and limitations visible.
Pilot completion	Evidence of seven days, 10-20 endpoints and two or more access media, if achieved.

Timeline is a proposed sequence, not a claim that the work or pilot has already been completed.

References & Methodology

Primary sources supporting the problem framing, measurement principles and submission requirements.

[1] FRAMEWORK FOR IP PERFORMANCE METRICS
Paxson, V., Almes, G., Mahdavi, J., & Mathis, M. (1998). RFC 2330, Internet Engineering Task Force.
https://www.rfc-editor.org/rfc/rfc2330.html

[2] OFFICIAL PROBLEM STATEMENT PS-014
AIORI-3 Hackathon. Pilot Internet-Quality Comparison by Verified Access Medium. Official problem statement, 2026.
https://drive.google.com/file/d/1VCIgJSDdSaXUQ09aZqpMjJ28RbF2cy1v/view?usp=drivesdk

[3] AIORI-3 HACKATHON GUIDELINES
AIORI. Official submission guidance and evaluation parameters, 2026.
https://portal.aiori.in/aiori-3/hackathon-guideline/

[4] OFFICIAL TEMPLATE REPOSITORY
AIORI-3 Hackathon. Proposed presentation structure and code repository guidelines.
https://github.com/aiori-hackathon/A3-PSXXX-TCXXX

HOW RFC 2330 IS USED

RFC 2330 provides a framework for thinking about metric definitions, measurement methodology and comparability. This proposal does not claim formal certification or complete compliance; the actual probes and procedures must be specified and validated.

CLAIM DISCIPLINE

This submission does not assert measured improvements, accuracy percentages, completed pilot results or deployed capabilities. Such claims should be added only when supported by reproducible measurements and demo evidence.

Conclusion & Expected Outcomes

A useful pilot is one that makes its evidence and its limitations equally visible.

The PS-014 proposal moves beyond a single headline download-speed result. It combines scheduled measurements of latency, packet loss, DNS resolution delay and bounded throughput with access-medium provenance, ISP/location context and local-link notes. The resulting dataset and dashboard are intended to show how observed quality varies across participating endpoints and time, while preserving the sample counts and limitations behind each comparison.

EXPECTED OUTPUTS

- Measurement agent
- Controlled measurement endpoint
- Labelled dataset
- Interactive dashboard
- Sample-count and uncertainty analysis
- Final report and demo

LIMITS TO RESPECT

- No national coverage claim
- No medium identification from IP alone
- No perfect Wi-Fi/ISP isolation claim
- No universal causal ranking
- No fabricated pilot results
- No unsupported accuracy score

SUCCESS CRITERIA

REPEATABLE
Same defined probes and schedule.

TRACEABLE
Timestamp, status and agent version retained.

CONTEXT-AWARE
Medium source and local context recorded.

HONEST
Uncertainty and limitations remain visible.

MEASURE THE CONNECTION, NOT JUST THE SPEED.

AIORI-3 PS-014 • A3-PS014-TC182

Team: Abhishek Giri • Ansh Kapoor | **Faculty Mentor:** Er. Mukesh Verma